

Bryan Emmanuel Alvarez Serna

Ph.D. in Electrical Engineering – Instrumentation

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Research Interests

- Electrical and electrochemical sensors.
- Chemical instrumentation.
- Environmental monitoring and water quality assessment.
- Biomedical sensing technologies.

Education

Ph.D. in Electrical Engineering – Instrumentation National Autonomous University of Mexico (UNAM), Mexico Graduated with Honors.	Jan. 2022 – Feb. 2026
M.Sc. in Electrical Engineering – Instrumentation National Autonomous University of Mexico (UNAM), Mexico Graduated with Honors.	Aug. 2019 – Oct. 2021
B.Sc. in Electrical and Electronic Engineering National Autonomous University of Mexico (UNAM), Mexico.	Aug. 2014 – Jul. 2019

Academic Appointments

Metropolitan Autonomous University (UAM), Lerma Campus Full-Time Visiting Professor, Titular C.	May 2026 – Present
Metropolitan Autonomous University (UAM), Lerma Campus Full-Time Associate Research Professor, Category D.	May 2025 – Apr. 2026
National Autonomous University of Mexico (UNAM), Faculty of Engineering Lecturer in Digital Design.	Aug. 2021 – May 2025

Research Experience

Institute of Physical Chemistry of Materials, Environment and Energy (INQUIMAE), University of Buenos Aires, Argentina Research stay focused on electrochemical sensors modified with polymers for heavy-metal detection in water. Supervisor: Dr. Lucila Paula Mendez De Leo.	Sep. 2024 – Dec. 2024
Arizona State University, USA Summer Research Internship. Supervisor: Prof. Jennifer Blain Christen.	Summer 2021

Selected Peer-Reviewed Journal Publications

1. **Alvarez-Serna, B. E.**, Rodriguez-Giron, J. S., Arzate-Salgado, S., Sanchez-Martinez, D., Ramirez-Chavarria, R. G., and Ramirez-Zamora, R. M. "Continuous Monitoring of the Photocatalytic Degradation of Levofloxacin Using an Impedimetric Sensor-Based System." *Processes*, 14(17), 2742, 2026.
2. Rodríguez-Girón, J. S., **Alvarez-Serna, B. E.**, Hernández-Uresti, D. B., Sánchez-Martínez, D., Gines-Palestino, R. S., Velázquez-Herrera, L. A., ... & Cabal-Prieto, A. 'Enhanced Hydrogen Production via Photocatalysis Using g-C₃N₄/ZIF-67 Hybrid Composites.' *Processes*, 14(2), 219, 2026.
3. Ramirez-Chavarria, R. G., Uc-Martin, J. A., **Alvarez-Serna, B. E.**, and Padilla-Moran, R. F. "Resonance-Induced Capacitively Coupled Contactless Conductivity Detection (ReC4D) Unit for Nucleic Acid Amplification Testing." *Technologies*, 13(4), 138, 2025.
4. **Alvarez-Serna, B. E.**, Arcos-Santiago, D. A., Uc-Martin, J. A., and Ramirez-Chavarria, R. G. "Electrochemical Sensor with Dynamic Self-Calibration for Acetaminophen Detection in Water." *IEEE Sensors Letters*, 2025.

5. **Alvarez-Serna, B. E.**, Garcia-Mejia, T. A., Arzate, S., Yanez-Aulestia, A., Ramirez-Chavarria, R. G., and Ramirez-Zamora, R. M. "Cost-effective Amperometric Sensor for Monitoring Levofloxacin in Groundwater." *Electroanalysis*, e202400096, 2024.
6. Ramirez-Chavarria, R. G., Castillo-Villanueva, E., **Alvarez-Serna, B. E.**, Carrillo-Reyes, J., Torres, L., Ramirez-Zamora, R. M., and Alvarez-Icaza, L. "Automatic Analysis of Isothermal Amplification via Impedance Time-Constant-Domain Spectroscopy: A SARS-CoV-2 Case Study." *Chemosensors*, 11(4), 230, 2023.
7. **Alvarez-Serna, B. E.**, Ramirez-Chavarria, R. G., Castillo-Villanueva, E., Carrillo-Reyes, J., Ramirez-Zamora, R. M., Buitron, G., and Alvarez-Icaza, L. "Label-free and Portable Field-Effect Sensor for Monitoring RT-LAMP Products to Detect SARS-CoV-2 in Wastewater." *Talanta*, 253, 124060, 2023.
8. Ramirez-Chavarria, R. G., Castillo-Villanueva, E., **Alvarez-Serna, B. E.**, Carrillo-Reyes, J., Ramirez-Zamora, R. M., Buitron, G., and Alvarez-Icaza, L. "Loop-Mediated Isothermal Amplification-Based Electrochemical Sensor for Detecting SARS-CoV-2 in Wastewater Samples." *Journal of Environmental Chemical Engineering*, 10(3), 107488, 2022.
9. Ramirez-Chavarria, R. G., **Alvarez-Serna, B. E.**, Schoukens, M., and Alvarez-Icaza, L. "Data-Driven Modeling of Impedance Biosensors: A Subspace Approach." *Measurement Science and Technology*, 32(10), 104009, 2021.

Complete publication record available through the [Google Scholar profile](#).

Selected Conference Publications

1. **Alvarez-Serna, B. E.** and Ramirez-Chavarria, R. G. "An IoT Electrochemical Sensor Based on a Conductive Polymer-Modified Electrode for Levofloxacin Determination." *Proceedings of the XIII Ibero-American Congress of Sensors (Ibersensor)*, 2024.
2. **Alvarez-Serna, B. E.**, Balderas-Zempoaltecatl, A. E., and Ramirez-Chavarria, R. G. "Molecularly Imprinted Polymer Paper-Based Biosensor for Wireless Measurement of Sweat Glucose." *XLV Mexican Conference on Biomedical Engineering*, Springer, pp. 611–618, 2022.
3. **Alvarez-Serna, B. E.** and Ramirez-Chavarria, R. G. "EGFET-Based pH Sensor Coupled with Low-Cost Electrochemical Screen-Printed Electrodes." *Journal of Physics: Conference Series*, 1723(1), 012024, 2021.

Awards and Distinctions

- Member of the Mexican National System of Researchers (*Sistema Nacional de Investigadoras e Investigadores*, SNII), Level I.
- ANUIES Award 2026, Doctoral Thesis category in Strategic Sectors for Mexico – Water.
- First Place, Scientific Challenge, IV Latin American Conference on Bioimpedance (CLABIO), 2021.
- Second Place, National Graduate Student Competition, XLV Mexican Conference on Biomedical Engineering (CNIB), 2022.

Patent

Inventor, Mexican Patent No. 426531:

Device and Electrochemical Method for Detecting SARS-CoV-2 Genetic Fragments in Wastewater.

Mexican Institute of Industrial Property (IMPI), granted 2025.

Scientific Leadership and Academic Service

- **President of the Scientific Committee**, V Congress on Mechatronic and Computational Systems, Metropolitan Autonomous University, 2026.
- **President of the Organizing Committee**, IV Congress on Mechatronic and Computational Systems, Metropolitan Autonomous University, 2025.
- Research seminars and invited academic talks on electrochemical sensors, water-quality monitoring, IoT-connected sensors, electronic instrumentation, and automation.
- Experience in undergraduate teaching in electrical circuits, electronic devices, digital design, VLSI design, automation and control, and scientific communication.
- Supervision and co-supervision of undergraduate research projects involving instrumentation, autonomous systems, data analysis, and environmental monitoring.

Peer-Review Activities

Reviewer for international peer-reviewed scientific journals in the fields of sensors, electrochemical systems, materials, and electronic instrumentation, including:

- *Talanta* and *Biosensors and Bioelectronics* (Elsevier).
- *Ionics* (Springer Nature).